Theoretical Formulation: Graph-Constrained Trajectory Language Model (GCT-LM) for Arterial Soft Route Discovery

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1 Problem Formalization

Let the urban road network be a directed graph $G = (V, E)$, restricted to the **Largest Strongly Connected Component (LSCC)** of drivable roads in the SUMO network (`manhattan_clean.net.xml`).

We define the discrete vocabulary $\mathcal{V}$ as the set of all directed edge IDs in the LSCC:

$$\mathcal{V} = \{e_1, e_2, \ldots, e_{|\mathcal{V}|}\}, \quad |\mathcal{V}| \approx 5{,}000 \tag{1}$$

A historical trip is represented as a discrete sequence of road segments $\mathbf{e} = [e_1, e_2, \ldots, e_L]$ occurring within a discrete 30-minute time bin $b \in \{0, 1, \ldots, 47\}$.

The objective is to discover a set of continuous, high-density arterial corridors (**Soft Routes**) $\mathcal{T}_b = \{\text{Trunk}_1, \ldots, \text{Trunk}_K\}$ for each time bin b, such that each corridor represents a high-probability directed traffic backbone without requiring rigid whole-path matching.

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2 Input Representation and Multi-Modal Embeddings

At each trajectory step $t \in \{1, \ldots, L\}$, the input vector $\mathbf{h}_t^{(0)} \in \mathbb{R}^d$ ($d = 256$) is constructed via the additive fusion of three continuous embedding manifolds:

$$\mathbf{h}_t^{(0)} = \mathbf{E}_{\text{edge}}(e_t) + \mathbf{E}_{\text{time}}(b) + \mathbf{E}_{\text{pos}}(t) \tag{2}$$

1. **Edge Token Embedding $\mathbf{E}_{\text{edge}} \in \mathbb{R}^{|\mathcal{V}| \times d}$:** Maps discrete SUMO edge IDs into a continuous latent space where co-occurring and spatially proximate roads project closely together.

2. **Continuous Temporal Embedding $\mathbf{E}_{\text{time}} \in \mathbb{R}^{48 \times d}$:** Encodes the 30-minute time bin $b \in \{0, \ldots, 47\}$. Unlike isolated bins, temporal representations are continuous and share parameter updates across adjacent hours. For harmonic index $k \in \{1, \ldots, d/2\}$:

$$\mathbf{E}_{\text{time}}^{(0)}(b)_{2k} = \sin\left(\frac{2\pi \cdot k \cdot b}{48}\right), \quad \mathbf{E}_{\text{time}}^{(0)}(b)_{2k+1} = \cos\left(\frac{2\pi \cdot k \cdot b}{48}\right) \tag{3}$$

   This mathematical continuity prevents off-peak collapse: low-demand hours (e.g., 3:30 AM) borrow statistical flow patterns from surrounding hours.

3. **Learnable Positional Encoding $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{L_{\max} \times d}$:** Preserves sequence order and flow directionality along the trip.

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3 Causal Attention and Topological Graph Masking

The sequence $\mathbf{H}^{(0)} = [\mathbf{h}_1^{(0)}, \ldots, \mathbf{h}_L^{(0)}]$ passes through N causal Transformer layers ($N = 4$, 8 attention heads) to compute the hidden state $\mathbf{h}_t^{(N)}$. The output projection computes raw logits over the entire road vocabulary:

$$\mathbf{z}_t = \mathbf{W}_{\text{vocab}} \mathbf{h}_t^{(N)} + \mathbf{b}_{\text{vocab}} \quad \in \mathbb{R}^{|\mathcal{V}|} \tag{4}$$

3.1 Topological Adjacency Masking (Physical Drivability Guarantee)

To prevent the model from generating disconnected roads (hallucinations) that would crash SUMO with routing errors, we enforce a strict topological constraint directly derived from the network graph.

Let $\mathcal{N}^+(e_t)$ denote the set of physically valid outgoing edges from edge e_t in the SUMO network:

$$\mathcal{N}^+(e_t) = \{e' \in \mathcal{V} \mid \text{FromNode}(e') = \text{ToNode}(e_t)\} \tag{5}$$

We define the state-dependent graph transition mask $\mathbf{M}_{\text{topo}}(e_t) \in \mathbb{R}^{|\mathcal{V}|}$ as:

$$(\mathbf{M}_{\text{topo}}(e_t))_j = \begin{cases} 0 & \text{if } j \in \mathcal{N}^+(e_t) \\ -\infty & \text{if } j \notin \mathcal{N}^+(e_t) \end{cases} \tag{6}$$

The constrained next-edge probability distribution is:

$$P(e_{t+1} = j \mid e_{\leq t}, b) = \frac{\exp\left((\mathbf{z}_t)_j + (\mathbf{M}_{\text{topo}}(e_t))_j\right)}{\sum_{k \in \mathcal{V}} \exp\left((\mathbf{z}_t)_k + (\mathbf{M}_{\text{topo}}(e_t))_k\right)} \tag{7}$$

Mathematical Guarantee: Because $\exp(-\infty) = 0$:

$$\forall j \notin \mathcal{N}^+(e_t), \quad P(e_{t+1} = j \mid e_{\leq t}, b) \equiv 0 \tag{8}$$

The model is mathematically incapable of predicting an illegal turn or disconnected edge. Every generated sequence is 100% continuous and drivable in SUMO.

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4 Training Objective and Corridor Extraction

4.1 Self-Supervised Objective

The model is trained offline on historical trip sequences using the Next-Token Prediction negative log-likelihood loss:

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(\mathbf{e}, b) \in \mathcal{D}} \sum_{t=1}^{L-1} \log P_\theta(e_{t+1} \mid e_1, \ldots, e_t, b) \tag{9}$$

4.2 Maximum A Posteriori (MAP) Beam Search Decoding

To extract soft routes for time bin b, we query the model from candidate arterial entry seeds $e_1 \in \mathcal{V}_{\text{entry}}$ (major incoming avenues, tunnel/bridge exits, southern tip gateways) using **Graph-Constrained Beam Search** ($W = 5$):

$$\text{Trunk}^* = \arg\max_{e_1, \ldots, e_m} \frac{1}{m^\gamma} \sum_{t=1}^{m-1} \log P(e_{t+1} \mid e_{\leq t}, b) \tag{10}$$

$$\text{subject to:} \quad e_{t+1} \in \mathcal{N}^+(e_t), \quad \sum_{t=1}^{m} \text{Length}(e_t) \geq 1.5 \text{ km}, \quad e_i \neq e_j \text{ (loop-free)}$$

where $\gamma = 0.7$ is a length normalization penalty.

Diversity Filtering: Pairwise Jaccard overlap between candidate trunks is enforced:

$$J(\text{TK}_a, \text{TK}_b) = \frac{|\text{TK}_a \cap \text{TK}_b|}{|\text{TK}_a \cup \text{TK}_b|} \leq 0.70 \tag{11}$$

Duplicate or near-identical parallel lanes are pruned, retaining only distinct arterial spines.

Simulation Output: Final corridors are exported directly to `dynamic_soft_routes.json`, requiring **zero modifications** to the downstream simulation engine (`03_run_ridesharing.py`).